Notes: Loutskina and Strahan (RFS, 2011)

Informed and Uninformed Investment in Housing: The Downside of Diversification

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1 Summary

Loutskina and Strahan (RFS, 2011) study how geographic diversification changes mortgage lenders' incentives to collect private information. The paper compares geographically concentrated lenders, which originate most of their loans in a small number of local markets, with diversified lenders, which operate across many markets and rely more heavily on scalable, public-information lending technologies.

The main hypothesis is that concentrated lenders have stronger incentives and better ability to collect local private information. They can therefore underwrite information-intensive borrowers and loans more effectively. Diversified lenders, by contrast, can scale lending more easily by conditioning on hard, verifiable signals such as FICO scores, loan-to-value ratios, and other information useful for securitization.

The paper finds that concentrated lenders behave as informed lenders. They are more active in the jumbo segment, accept more applications, retain more loans on their balance sheets, and lend to riskier observable borrowers, especially in their primary markets. Concentrated lenders also earn higher profits, have profits that are less exposed to national house-price cycles, and experience better stock returns during the 2007–2008 credit crisis.

The broader implication is that the rise of diversified mortgage lending reduced screening and monitoring in housing finance. This decline in private-information production likely contributed to the housing boom and the subsequent crisis.

2 Big picture

- **Question.** Does geographic diversification reduce lenders' incentives to produce private information in mortgage markets?
- **Core comparison.** Concentrated lenders versus diversified lenders.
- **Key mechanism.** Private information is costly and local. Concentrated lenders have a comparative advantage in local screening, but this information makes their loans less liquid because outside buyers worry about adverse selection.
- **Empirical setting.** U.S. mortgage lending from 1992 to 2007, using HMDA mortgage applications, bank regulatory data, and bank stock returns during the crisis.
- **Identification idea.** Compare lender behavior across the jumbo cutoff, across lenders with different geographic concentration, and within concentrated lenders' primary versus satellite markets.
- **Main conclusion.** Diversification helped create scalable lending but reduced private screening, especially as concentrated lenders' market share fell during the housing boom.

3 Economic mechanism

The paper contrasts two lending technologies.

Concentrated-lender technology. A concentrated lender specializes in one or a few local markets. It can collect soft and local information about borrowers, properties, employers, neighborhoods, and local real estate conditions. This private information helps the lender price risk and approve loans that a less informed lender might reject. However, the same information makes the lender's loan portfolio more difficult to sell: outside buyers cannot verify the lender's private signal, so adverse selection reduces loan liquidity.

Diversified-lender technology. A diversified lender operates across many markets. It relies more on public and verifiable information, such as FICO scores, loan-to-value ratios, income, and other variables that are easier to process at scale and easier for secondary-market investors or rating agencies to verify. This technology supports loan sales and securitization, but it can sacrifice local screening.

The model intuition is straightforward. Private information should lead to higher acceptance rates because informed lenders can distinguish good risky borrowers from bad ones. It should also lead to higher loan retention rates because loans originated using private information are less liquid. If screening is valuable, concentrated lenders should earn higher average profits and perform better during a housing downturn, despite serving riskier borrowers.

4 Empirical design

4.1 Measuring lender concentration

The paper measures a lender's geographic concentration with a Herfindahl index of its mortgage lending across metropolitan statistical areas (MSAs). Lenders with an HHI above 0.50 are classified as concentrated lenders in the descriptive figures. Regression specifications use continuous concentration measures based on application counts and dollar volume.

Figure 1 documents the central time-series fact. From 1992 to 2007, national house prices rose sharply, while the market share of concentrated lenders fell dramatically. This is the basic macro pattern motivating the paper: the housing boom coincided with the growing dominance of diversified lending.

4.2 Jumbo versus non-jumbo loans

The jumbo cutoff is central because non-jumbo mortgages are more easily purchased or guaranteed by the government-sponsored enterprises, Fannie Mae and Freddie Mac. GSE underwriting depends mostly on public and verifiable signals. Jumbo loans are more heterogeneous and less subsidized by the GSEs, so they are more private-information intensive.

The paper therefore asks whether concentrated lenders tilt toward jumbo mortgages and whether the shift from non-jumbo to jumbo loans affects concentrated lenders less than diversified lenders. If concentrated lenders produce private information, they should be more willing to lend and retain loans in the jumbo segment.

4.3 Acceptance and retention tests

The paper estimates regressions for application acceptance rates and loan retention rates. A higher acceptance rate suggests that the lender can evaluate borrower risk more precisely and ration credit less. A higher retention rate suggests that the lender's private information makes loans harder to sell because buyers discount privately screened loans.

Figure 2 provides a visual version of the results around the jumbo cutoff. Concentrated lenders accept more applications and sell fewer loans than diversified lenders. The differences are especially visible around the size threshold that separates non-jumbo and jumbo loans.

4.4 Within-bank tests

The paper also compares concentrated lenders' behavior in their primary markets to their behavior in satellite markets. This test addresses the concern that concentration is just another proxy for bank size or business model. If local knowledge matters, the same concentrated bank should look more informed in its core market than in its noncore markets.

Table 5 implements this within-bank design. Concentrated lenders behave more like diversified lenders when operating outside their core markets: they make relatively more non-jumbo loans, accept fewer mortgages, and sell more originations.

5 Main findings

5.1 Concentrated lenders focus on more information-intensive lending

The paper finds that concentrated lenders allocate relatively more activity toward jumbo mortgages. This pattern is consistent with the idea that concentrated lenders have an advantage in segments where private information has greater value and where public-information underwriting is less sufficient.

5.2 Concentrated lenders ration credit less

Concentrated lenders have higher acceptance rates after controlling for borrower and bank characteristics. This result is consistent with informed underwriting: these lenders are better able to separate good applications from bad ones and therefore reject fewer applications for a given level of observable borrower risk.

5.3 Concentrated lenders retain more loans

Concentrated lenders sell fewer of their originated mortgages and retain more loans on balance sheet. This is consistent with adverse selection in secondary markets. If a lender originates based on private information, outside buyers cannot fully verify the quality of retained versus sold loans, making private-information loans less liquid.

5.4 Concentration is not simply bank size

The within-bank results show that concentrated lenders act differently in their primary markets than in satellite markets. This is an important validation of the paper's interpretation. The local-information channel predicts exactly this pattern: the lender's advantage should be strongest where it has local expertise.

5.5 Concentrated lenders are profitable and resilient

The paper finds that concentrated lenders earn higher accounting profits, and their profits are less exposed to aggregate housing-market conditions. Concentrated lenders also have better stock-market performance during the crisis period from August 2007 to March 2008. These facts reduce the concern that higher acceptance rates simply reflect weaker credit standards.

5.6 Concentrated lenders serve riskier borrowers

Table 8 shows that concentrated lenders' applicant pools are observably riskier, especially in primary markets. This result reinforces the central interpretation. Concentrated lenders are not safer simply because they avoid risky borrowers; rather, their screening allows them to lend to riskier pools while maintaining better outcomes.

6 How to read the paper's main exhibits

- **Figure 1** shows the rise in house prices and the decline in the market share of concentrated lenders.
- **Figure 2** compares acceptance and loan-sale rates around the jumbo cutoff for concentrated and diversified lenders.
- **Table 1** provides summary statistics for bank characteristics and mortgage behavior.
- **Tables 2–4** are the core lending-behavior regressions: jumbo activity, acceptance rates, and retention rates.
- **Table 5** implements the within-bank primary-market test.
- **Table 6** studies profitability and nonperforming loans.
- **Table 7** studies bank stock returns during the crisis.
- **Table 8** shows that concentrated lenders serve riskier borrower pools in their primary markets.
- **Figure 3** relates early lender concentration to subsequent house-price appreciation across Case-Shiller MSAs.

7 Contribution and interpretation

The paper links two major developments in U.S. mortgage markets: geographic diversification and securitization. Diversification created scalable lending institutions, while securitization rewarded hard-information loan production. Together, these forces reduced the relative importance of locally informed lenders.

The paper's contribution is not simply that concentrated lenders behave differently. It shows that the differences line up across multiple margins: product choice, acceptance, retention, profitability, crisis performance, and borrower risk. This pattern supports a coherent interpretation in which local concentration increases private-information production.

The implication for the financial crisis is subtle. Diversification can reduce idiosyncratic geographic risk, but it can also reduce screening. If diversified lenders rely too much on public signals, aggregate mortgage credit may expand with less private monitoring. The paper therefore highlights a downside of diversification: a safer-looking, more scalable lending system may produce less information and worse screening.

8 Extracted figures and tables from the paper

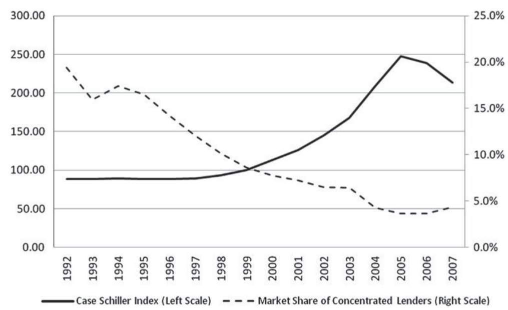

Figure 1
Trends in house prices and market share of concentrated lenders
This figure reports the Case-Shiller index for national housing prices over time, along with the dollar-weighted fraction of all mortgage lending among institutions with an HHI index of lending across metropolitan statistical areas of at least 0.50 (concentrated lenders).

Figure 1. Trends in house prices and the market share of concentrated lenders.

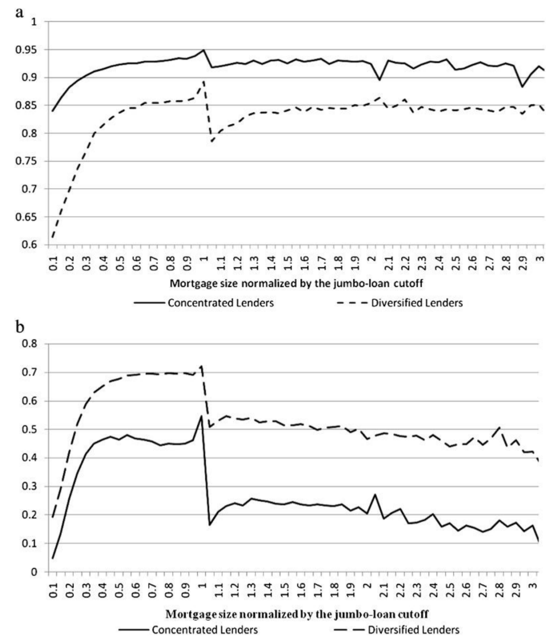

Figure 2
A) Probability of acceptance by mortgage size B) Share of mortgages sold, by size
These figures plot the average acceptance rates (Panel A) and retention rates (Panel B) for mortgages against the ratio of the size of the mortgage scaled by the jumbo-mortgage cutoff. The data are from HMDA and cover the 1992 to 2007 sample period, and are based on the authors' calculations. Concentrated lenders are those with an HHI index of lending across metropolitan statistical areas of at least 0.50. Diversified lenders are those offering

Figure 2. Acceptance rates and mortgage sales by loan size.

Table 1
Summary statistics for bank characteristics

	Panel A: All Banks		*Panel B: Concentrated Banks*		*Panel C: Diversified Banks*	
	Mean	St. Dev.	Mean	St. Dev.	Mean	St. Dev.
Total Assets (millions of $s)	391.1	6661.4	170.0	3355.1	2474.1	18750.0
Financial Struture & Liquidity (%)						
Liquid assets/assets	30.5	15.6	31.2	15.6	23.9	13.7
Cost of deposits	3.2	5.9	3.3	6.2	2.7	1.1
Capital/assets	10.1	4.1	10.2	4.2	9.4	3.2
Loan Shares (% of assets)						
Total loans	59.5	15.3	58.8	15.3	65.8	13.8
Commercial & industrial loans	22.2	18.7	23.2	18.9	13.1	14.2
Home mortgages	36.6	30.0	36.6	29.6	36.8	30.4
Commercial mortgages	17.1	13.2	16.4	12.9	23.5	14.3
Consumer loans	13.6	11.7	14.0	11.7	9.7	11.0
Profit (%)						
Net income/assets	0.90	1.13	0.90	1.14	0.95	0.98
Herfindahl Index						
Equal weighted	0.841	0.227	0.959	0.065	0.495	0.176
Value weighted	0.834	0.231	0.948	0.092	0.498	0.189
Mortgage application						
Number of loan applications	1810	23326	274	829	5444	42565
Number of loans issued	1457	19909	247	762	4321	36355
Number of loans securitized	775	13393	88	534	2400	24487
Probability of acceptance (%)	87.53	13.33	88.06	12.60	86.31	14.80
Share of loans retained (%)	79.20	31.81	81.76	30.48	73.26	33.97
Average loan amount ($000)	134	350	113	178	186	578
Average applicant income ($000)	95	127	88	99	112	174
Average loan-to-income ratio	1.67	3.08	1.61	3.38	1.81	2.20
Averate loan to area income	2.07	2.41	2.00	2.02	2.24	3.15
Percent minority	16.40	15.80	15.49	15.89	18.57	15.36
Percent female	17.31	10.80	16.70	10.99	18.77	10.18

This table reports information on the distribution of characteristics for banks matched to the mortgage application data. Bank financial information is from Call Reports, while the loan applications and origination data are from HMDA. The sample covers the period from 1992 to 2007. Concentrated lenders are those with HHI index of lending across metropolitan statistical areas of at least 0.50, and diversified lenders are those making mortgages

Table 1. Summary statistics for bank characteristics.

Table 2
Regression of loan volumes for non-jumbo mortgages relative to jumbos on loan concentration

	Dependent Variable: (Volume of Approved Non−jumbos − Volume of Jumbos)/Assets$_{t-1}$					
	Full Sample		Prior to 2001		Post 2001	
	(1)	(2)	(3)	(4)	(5)	(6)
Loan concentration (number)	−0.147***	–	−0.111***	–	−0.192***	–
	(4.808)	–	(2.793)	–	(4.485)	–
Loan concentration ($ volume)	–	−0.133***	–	−0.104***	–	−0.168***
	–	(4.620)	–	(2.765)	–	(4.137)
Securities/assets	−0.128***	−0.132***	−0.0901***	−0.0928***	−0.179***	−0.186***
	(5.121)	(5.234)	(4.306)	(4.397)	(4.358)	(4.464)
Interest on deposits/deposits	−0.611**	−0.600**	−0.371	−0.368	−0.971	−0.945
	(2.284)	(2.245)	(1.558)	(1.551)	(1.611)	(1.565)
Log of assets	−0.0375***	−0.0364***	−0.0306***	−0.0301***	−0.0474***	−0.0452***
	(7.383)	(7.336)	(7.017)	(7.131)	(4.622)	(4.472)
Indicator if owned by BHC	−0.00479	−0.00496	0.00423	0.00426	−0.0233***	−0.0238***
	(0.985)	(1.021)	(0.961)	(0.967)	(2.657)	(2.715)
Capital/assets	−0.187***	−0.183***	−0.114	−0.112	−0.246**	−0.241**
	(2.968)	(2.912)	(1.187)	(1.165)	(2.387)	(2.324)
Deposits/assets	−0.287***	−0.290***	−0.198***	−0.199***	−0.387***	−0.391***
	(3.604)	(3.622)	(3.450)	(3.464)	(2.890)	(2.901)
Net income/assets	0.724	0.702	0.22	0.207	1.675	1.629
	(1.375)	(1.331)	(1.086)	(1.027)	(1.198)	(1.162)
Real estate loans/assets	0.122***	0.118***	0.152***	0.150***	0.0845	0.0782
	(3.812)	(3.699)	(5.782)	(5.776)	(1.565)	(1.438)
C&I loans/assets	−0.181***	−0.183***	−0.126***	−0.128***	−0.273***	−0.274***
	(5.273)	(5.325)	(4.801)	(4.879)	(4.099)	(4.102)
Unused loan commitments/assets	−0.000991	−0.000947	−0.000317	−0.000276	0.00141	0.00133
	(0.580)	(0.555)	(0.252)	(0.218)	(0.348)	(0.331)

(*continued*)

Table 2, part 1. Non-jumbo versus jumbo loan volumes and lender concentration.

Table 2
Continued

	Dependent Variable: (Volume of Approved Non−jumbos − Volume of Jumbos)/Assets_{t-1}					
	Full Sample		Prior to 2001		Post 2001	
	(1)	(2)	(3)	(4)	(5)	(6)
Letters of credit/assets	−0.241*	−0.286**	−0.162	−0.188	−0.299	−0.372*
	(1.775)	(2.075)	(0.801)	(0.954)	(1.471)	(1.774)
Observations	68,291	68,291	40,326	40,326	27,965	27,965
R−squared	7.7%	7.6%	6.8%	6.7%	9.4%	9.1%

This table reports regressions of the volume of approved non-jumbo mortgages minus jumbo mortgages, divided by beginning-of-period assets. The unit of observation is the bank-year from 1992 to 2007. The regressions include the following controls for the loan pool characteristics in each bank-year: percentage of minority applicants in the bank's lending markets, mean loan-to-income ratio, log of mean applicant income, average median income in the bank's lending markets, proportion of minority applicants, and proportion of female applicants. All regressions also include the unemployment rate, personal income growth, the percentage of the population older than 65, and the log of population density (each based on averages across MSA markets weighted by the number of loans originated), as well as year and state fixed effects. *t*-statistics are in parentheses and are based on errors clustered at the bank level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 2, continued. Loan-volume regressions and table notes.

Table 3
Regression of acceptance rates for mortgages on loan concentration

	Dependent Variable: Number of Approved Mortgages/ Number of Applications					
	Full Sample		Prior to 2001		Post 2001	
	(1)	(2)	(3)	(4)	(5)	(6)
Loan concentration (number)	0.0377***	0.0323***	0.0387***	0.0375***	0.0356***	0.0266***
	(7.679)	(6.348)	(6.185)	(5.689)	(5.905)	(4.259)
Jumbo * loan concentration	–	0.0139***	–	0.0123***	–	0.0238***
	–	(2.995)	–	(2.503)	–	(4.131)
Jumbo loan indicator	−0.00219	−0.0132***	−0.00209	−0.00479	−0.00504**	−0.0228***
	(1.147)	(3.040)	(0.632)	(0.715)	(2.257)	(4.541)
Securities/assets	0.0220**	0.0220**	0.0610***	0.0610***	−0.0225*	−0.0225*
	(2.427)	(2.429)	(4.901)	(4.899)	(1.931)	(1.932)
Interest on deposits/deposits	0.157	0.162	0.109	0.111	0.11	0.112
	(1.153)	(1.192)	(0.788)	(0.801)	(0.493)	(0.505)
Log of assets	−0.0014	−0.00144	−0.0013	−0.00131	−0.0019	−0.00195
	(1.387)	(1.428)	(1.186)	(1.198)	(1.218)	(1.251)
Indicator if owned by BHC	−0.00104	−0.00104	0.00174	0.00174	−0.00654**	−0.00653**
	(0.491)	(0.491)	(0.691)	(0.692)	(2.265)	(2.261)
Capital/assets	0.0632**	0.0636**	0.0944***	0.0944***	−0.0185	−0.0173
	(2.472)	(2.490)	(2.785)	(2.787)	(0.567)	(0.531)
Net income/assets	0.386***	0.388***	0.375***	0.376***	0.324*	0.327*
	(3.441)	(3.459)	(2.778)	(2.781)	(1.767)	(1.787)
Deposits/assets	−0.00447	−0.00473	−0.0233	−0.0234	−0.00293	−0.00324
	(0.277)	(0.295)	(1.372)	(1.377)	(0.153)	(0.169)
Real estate loans/assets	0.0648***	0.0641***	0.0897***	0.0896***	0.0450***	0.0438***
	(6.299)	(6.228)	(6.590)	(6.563)	(3.625)	(3.523)
C&I loans/assets	0.0556***	0.0554***	0.0880***	0.0879***	0.0390**	0.0386**
	(4.094)	(4.076)	(4.910)	(4.906)	(2.287)	(2.271)
Unused loan commitments/ assets	−0.00333***	−0.00334***	−0.00277***	−0.00277***	−0.0347*	−0.0350*
	(3.764)	(3.766)	(6.372)	(6.374)	(1.792)	(1.807)

(*continued*)

Table 3, part 1. Acceptance rates and lender concentration.

Table 3
Continued

	Dependent Variable: Number of Approved Mortgages/ Number of Applications					
	Full Sample		Prior to 2001		Post 2001	
	(1)	(2)	(3)	(4)	(5)	(6)
Letters of credit/assets	0.0733	0.0719	0.0558	0.0538	0.159	0.161
	(0.863)	(0.848)	(0.381)	(0.368)	(1.470)	(1.490)
Observations	94,582	94,582	54,620	54,620	39,962	39,962
R−squared	13%	13%	14%	14%	12%	12%

This table reports regressions of the acceptance rates for jumbo and non-jumbo loan applications by bank-year. There are two observations per bank-year from 1992 to 2007. The regressions include the following controls for the loan pool characteristics in each bank-year: percentage of minority applicants in the bank's lending markets, mean loan-to-income ratio, log of mean applicant income, average median income in the bank's lending markets, proportion of minority applicants, and proportion of female applicants. All regressions also include the unemployment rate, personal income growth, the percentage of the population older than 65, and the log of population density (each based on averages across MSA markets weighted by the number of loans originated), as well as year and state fixed effects. *t*-statistics are in parentheses and are based on errors clustered at the bank level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 3, continued. Acceptance-rate regressions and table notes.

Table 4
Regression of retention rate for mortgages on loan concentration

	Dependent Variable: Volume of Approved Mortgages Retained on Balance Sheet/ Volume of Approved Mortgages					
	Full Sample		Prior to 2001		Post 2001	
	(1)	(2)	(3)	(4)	(5)	(6)
Loan concentration (number)	0.119***	0.153***	0.0906***	0.125***	0.139***	0.171***
	(8.803)	(10.660)	(5.724)	(7.444)	(7.665)	(9.039)
Jumbo * loan concentration	–	−0.0875***	–	−0.0896***	–	−0.0842***
	–	(8.626)	–	(7.702)	–	(6.060)
Jumbo loan indicator	0.0936***	0.163***	0.0980***	0.173***	0.0880***	0.151***
	(22.160)	(17.820)	(16.490)	(14.910)	(16.810)	(12.930)
Securities/assets	0.208***	0.208***	0.207***	0.208***	0.212***	0.212***
	(7.786)	(7.792)	(6.220)	(6.247)	(6.027)	(6.031)
Interest on deposits/deposits	0.709**	0.676**	0.901***	0.856***	−0.225	−0.233
	(2.313)	(2.208)	(2.985)	(2.843)	(0.395)	(0.409)
Log of assets	−0.00204	−0.00181	0.00819***	0.00849***	−0.0216***	−0.0215***
	(0.735)	(0.650)	(2.801)	(2.901)	(4.939)	(4.904)
Indicator if owned by BHC	−0.0250***	−0.0250***	−0.0345***	−0.0345***	−0.0121	−0.0121
	(4.177)	(4.180)	(5.090)	(5.106)	(1.379)	(1.380)
Capital/assets	0.527***	0.524***	0.686***	0.684***	0.350***	0.346***
	(6.892)	(6.866)	(7.484)	(7.479)	(3.117)	(3.087)
Deposits/assets	0.00775	−0.00624	0.102	0.0864	−0.0974	−0.107
	(0.026)	(0.021)	(0.351)	(0.297)	(0.175)	(0.193)
Net income/assets	0.168***	0.169***	0.173***	0.174***	0.151***	0.152***
	(4.329)	(4.370)	(3.723)	(3.764)	(3.081)	(3.099)
Real estate loans/assets	0.0829***	0.0872***	0.0741**	0.0785**	0.0882**	0.0928**
	(2.834)	(2.982)	(2.065)	(2.189)	(2.375)	(2.497)
C&I loans/assets	0.0335	0.0352	0.0234	0.025	0.0667	0.0681
	(0.835)	(0.878)	(0.465)	(0.496)	(1.301)	(1.328)
Unused loan commitments/ assets	0.001	0.001	0.001	0.001	−0.0717***	−0.0707***
	(0.136)	(0.175)	(1.011)	(1.080)	(3.295)	(3.236)

(*continued*)

Table 4, part 1. Retention rates and lender concentration.

Table 4
Continued

	Dependent Variable: Volume of Approved Mortgages Retained on Balance Sheet/ Volume of Approved Mortgages					
	Full Sample		Prior to 2001		Post 2001	
	(1)	(2)	(3)	(4)	(5)	(6)
Letters of credit/assets	−0.655**	−0.646**	−2.012***	−1.954***	0.893***	0.888***
	(2.542)	(2.512)	(5.042)	(4.900)	(3.442)	(3.424)
Observations	92,921	92,921	53,555	53,555	39,366	39,366
R−squared	12%	12%	11%	11%	16%	16%

This table reports regressions of the retention rate for jumbo and non-jumbo loan applications by bank-year. There are two observations per bank-year from 1992 to 2007. The regressions include the following controls for the loan pool characteristics in each bank-year: percentage of minority applicants in the bank's lending markets, mean loan-to-income ratio, log of mean applicant income, average median income in bank's lending markets, proportion of minority applicants, and the share of female applicants. All regressions also include the unemployment rate, personal income growth, the percentage of the population older than 65, and the log of population density (each based on averages across MSA markets weighted by the number of loans originated), as well as year and state fixed effects. *t*-statistics are in parentheses and are based on errors clustered at the bank level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 4, continued. Retention-rate regressions and table notes.

Table 5
Within-bank results

	(Volume of Approved Non-jumbos - Volume of Jumbos$_t$)/ Total Volume$_{t-1}$	Number of Approved Mortgages/ Number of Applications		Volume of Approved Mortgages Retained on Balance Sheet/ Volume of Approved Mortgages	
	(1)	(2)	(3)	(4)	(5)
Primary market indicator	−0.244***	0.0420***	0.0500***	0.0311***	0.0396***
	(24.87)	(48.99)	(51.84)	(19.76)	(22.37)
Jumbo * primary market	–	–	0.0290***	–	−0.0314***
	–	–	(18.01)	–	(10.50)
Jumbo loan indicator	–	0.000268	−0.0172***	0.0940***	0.113***
	–	(0.30)	(13.05)	(56.53)	(45.99)
Securities/assets	0.0294	−0.0164**	−0.0152**	0.0614***	0.0602***
	(0.36)	(2.17)	(2.02)	(4.42)	(4.33)
Interest on deposits/ deposits	−1.059	−0.010	−0.012	0.366**	0.367**
	(1.17)	(0.13)	(0.15)	(2.47)	(2.48)
Log of assets	−0.0903***	−0.00246	−0.00253	−0.0153***	−0.0152***
	(4.92)	(1.53)	(1.58)	(5.18)	(5.14)
Indicator if owned by BHC	−0.0209	0.000735	0.000859	−0.0111***	−0.0112***
	(0.96)	(0.37)	(0.43)	(3.02)	(3.05)
Capital/assets	0.925***	0.0167	0.0161	0.247***	0.247***
	(3.07)	(0.74)	(0.71)	(5.96)	(5.97)
Deposits/assets	0.308***	0.475***	0.482***	0.047	0.0385
	(2.82)	(6.43)	(6.53)	(0.35)	(0.28)
Net income/assets	0.893	0.00515	0.00368	0.121***	0.123***
	(0.92)	(0.52)	(0.37)	(6.63)	(6.72)
Real estate loans/assets	−0.847***	0.004	0.005	0.00111	0.000
	(9.29)	(0.50)	(0.56)	(0.07)	(0.03)
C&I loans/assets	−0.632***	0.0153	0.0162	−0.115***	−0.116***
	(5.63)	(1.52)	(1.62)	(6.23)	(6.29)
Unused loan commitments/assets	0.018	0.00840**	0.00837**	−0.000633	−0.001
	(0.35)	(2.47)	(2.46)	(0.10)	(0.10)
Letters of credit/assets	1.222**	−0.397***	−0.401***	−0.371***	−0.366***
	(2.57)	(9.99)	(10.12)	(5.10)	(5.02)
Observations	76,552	179,457	179,457	177,583	177,583
Number of Fixed Effects	7,190	7,977	7,977	7,977	7,977
R−squared (within)	4.8%	3.6%	3.7%	5.4%	5.5%

This table reports within-bank regressions of the relative business in non-jumbos, the retention rate, and the loan approval rate. For each bank-year, we construct three variables for each MSA in which the bank makes mortgages. The primary market indicator equals one for the market in which the bank has the highest share of loans. In columns (2)–(5), each of these is computed separately for non-jumbos and jumbos. For these tests, we include only banks with at least 75% of their loan applications in one market. The regressions include the following controls for the loan pool characteristics in each bank-year: percentage of minority applicants in the bank's lending markets, mean loan-to-income ratio, log of mean applicant income, average median income in bank's lending markets, proportion of minority applicants, and proportion of female applicants. All regressions also include MSA-level unemployment, personal income growth, the percentage of the population older over 65, and the log of population density, as well as year, bank, and MSA fixed effects. *t*-statistics are in parentheses and are based on errors clustered at the bank level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 5. Within-bank primary-market results.

Table 6
Regression of accounting profit rate on loan concentration and other bank characteristics

	ROA		ROE		NPL / Loans	
	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: Full Sample					
Loan concentration (number)	0.000734***	0.00114***	0.00740***	0.0117***	−0.00180***	−0.00421***
	(9.56)	(14.20)	(8.70)	(13.25)	(4.76)	(9.07)
Loan concentration (number)	–	−0.00565***	–	−0.0591***	–	0.0334***
* Growth in Case-Shiller Index	–	(10.79)	–	(10.14)	–	(10.78)
Observations	247,568	247,568	247,602	247,602	249,100	249,100
R−squared	27%	27%	30%	30%	7%	7%
	Panel B: Heavy Mortgage Lenders (originations / assets > median)					
Loan concentration (number)	0.000767***	0.00124***	0.00772***	0.0125***	−0.00171***	−0.00511***
	(7.69)	(11.53)	(6.97)	(10.52)	(3.48)	(7.92)
Loan concentration (number)	–	−0.00668***	–	−0.0678***	–	0.0478***
* Growth in Case-Shiller Index	–	(8.11)	–	(7.33)	–	(10.14)
Observations	124,332	124,332	124,120	124,120	124,342	124,342
R-quared	33%	33%	36%	36%	4%	4%

(*continued*)

Table 6, part 1. Accounting profitability, nonperforming loans, and concentration.

Table 6
Continued

	Panel C: Prior to 2001					
Loan concentration (number)	0.000770***	0.00130***	0.00130***	0.0134***	−0.00313***	−0.00599***
	(8.74)	(13.46)	(13.46)	(12.31)	(6.01)	(9.26)
Loan concentration (number)	–	−0.0129***	–	−0.144***	–	0.0691***
* Growth in Case-Shiller Index	–	(12.75)	–	(12.47)	–	(11.79)
Observations	136,803	136,803	136,498	136,498	136,803	136,803
R-squared	32%	32%	34%	34%	6%	6%
	Panel D: Post 2001					
Loan concentration (number)	0.000756***	0.000913***	0.00770***	0.00970***	−0.000789*	−0.0006
	(7.42)	(7.91)	(6.80)	(7.78)	(1.83)	(1.25)
Loan concentration (number)	–	−0.00153***	–	−0.0196***	–	−0.0017
* Growth in Case-Shiller Index	–	(2.82)	–	(3.41)	–	(0.68)
Observations	110,765	110,765	111,104	111,104	112,297	112,297
R−squared	28%	29%	31%	32%	5%	5%

This table reports regressions of bank quarterly ROA (income/assets), ROE (income/book value of equity), and NPL/loans (loans 90 or more days past due plus loans no longer accruing interest/total loans) by bank-quarter from 1992 to 2007. The regressions include the following controls for bank characteristics: log of assets, securities/assets, interest expenses/deposits, deposits/assets, real estate loans/assets, C&I loans/assets, unused commitment/assets, and letters of credit/assets. All regressions also include the log of population density faced by banks (based on a loan-weighted average across MSA markets), and year and state fixed effects. *t*-statistics are in parentheses and are based on errors clustered at the bank level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 6, continued. Subsample profitability results and table notes.

Table 7
Regression of stock return from August 2007 to March 2008 on bank characteristics

	Dependent Variable: Cumulative Stock Return			
	(1)	(2)	(3)	(4)
Loan concentration (number)	0.165	–	0.128	–
	(3.03)***	–	(2.04)**	–
Loan concentration ($ volume)	–	0.162	–	0.111
	–	(2.94)***	–	(1.80)*
Securities/assets	0.781	0.767	−0.051	−0.037
	(5.55)***	(5.47)***	(0.21)	(0.15)
Interest on deposits/deposits	–	–	−0.19	−0.191
	–	–	(3.24)***	(3.28)***
Log of assets	–	–	−0.012	−0.014
	–	–	(0.71)	(0.83)
Capital/assets	–	–	−1.123	−1.099
	–	–	(1.40)	(1.36)
Deposits/assets	–	–	−0.234	−0.23
	–	–	(1.05)	(1.03)
Net income/assets	–	–	−6.525	−6.44
	–	–	(1.20)	(1.16)
Real estate loans/assets	–	–	−0.653	−0.638
	–	–	(2.93)***	(2.86)***
C&I loans/assets	–	–	–0.26	−0.256
	–	–	(0.91)	(0.89)
Unused loan commitments/assets	–	–	−0.523	−0.52
	–	–	(2.09)**	(2.06)**
Letters of credit/assets	–	–	−0.703	−0.652
	–	–	(0.95)	(0.86)
Observations	313	313	313	313
R-squared	13%	13%	24%	24%

This table reports cross-sectional regressions of cumulative returns for publicly traded banks' stocks between August 2007 and March 2008. *t*-statistics are in parentheses and are based on robust standard errors. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

deviations resulted in an increase in stock returns of about 6%, which is nearly

Table 7. Crisis-period stock returns and bank characteristics.

Table 8
Concentrated banks' applicant risk pool is greater in primary markets

	Dependent Variable = Average applicant income / area income			
	Across Banks		Within−Bank	
	(1)	(2)	(3)	(4)
Loan concentration/Primary market indicator	−0.203**	−0.181**	−0.735***	−0.392***
	(2.044)	(1.984)	(23.85)	(11.32)
Jumbo * Loan Conc/Primary market indicator	–	−0.172***	–	−1.272***
	–	(3.449)	–	(21.660)
Jumbo loan indicator	1.891***	2.031***	3.517***	4.269***
	(61.720)	(20.400)	(119.80)	(93.910)
Securities/assets	−0.122	−0.122	−0.846***	−0.886***
	(0.574)	(0.573)	(3.26)	(3.422)
Interest on deposits/deposits	−5.766***	−5.802***	1.688	1.449
	(2.879)	(2.898)	(0.68)	(0.587)
Log of assets	0.138***	0.138***	0.408***	0.412***
	(7.007)	(7.012)	(7.80)	(7.883)
Indicator if owned by BHC	0.0883*	0.0883*	−0.0731	−0.0775
	(1.873)	(1.874)	(1.01)	(1.075)
Capital/assets	0.969	0.967	1.508*	1.542*
	(1.572)	(1.569)	(1.89)	(1.938)
Net income/assets	−6.206***	−6.229***	2.021	1.652
	(2.752)	(2.766)	(0.76)	(0.623)
Deposits/assets	−0.0193	−0.0175	0.248	0.305
	(0.065)	(0.059)	(0.72)	(0.887)
Real estate loans/assets	−0.555**	−0.551**	−0.45	−0.460
	(2.357)	(2.339)	(1.57)	(1.603)
C&I loans/assets	0.916**	0.919**	−0.687**	−0.733**
	(2.199)	(2.202)	(1.98)	(2.120)
Unused loan commitments/assets	0.002	0.002	−0.067	−0.066
	(0.302)	(0.304)	(0.55)	(0.540)
Letters of credit/assets	7.879***	7.885***	−2.33	−2.164
	(3.729)	(3.731)	(1.60)	(1.490)
Observations	100,247	100,294	194,449	194,449
Bank Fixed Effects	No	No	Yes	Yes
R−squared (within)	8.5%	8.5%	7.7%	7.9%

This table reports regressions of the average applicant-to-local-area-income ratio within and across banks. In columns (1) and (2), we construct the average applicant-to-area-income ratio separately for jumbo and non-jumbo loans for each bank-year. In columns (3) and (4), we construct the variables for each MSA in which the bank makes mortgages. The sample for columns (1) and (2) includes only loans between 50% and 250% of the jumbo-loan cutoff. For columns (3) and (4), we include only banks with at least 75% of their loan applications in one market. The regressions include the following MSA-year controls: unemployment, personal income growth, percentage of population older than 65 years old, and log population density. The regressions also include year, MSA, and (specifications (3) and (4)) bank fixed effects. *t*-statistics are in parentheses and are

Table 8. Applicant risk pools in primary markets.

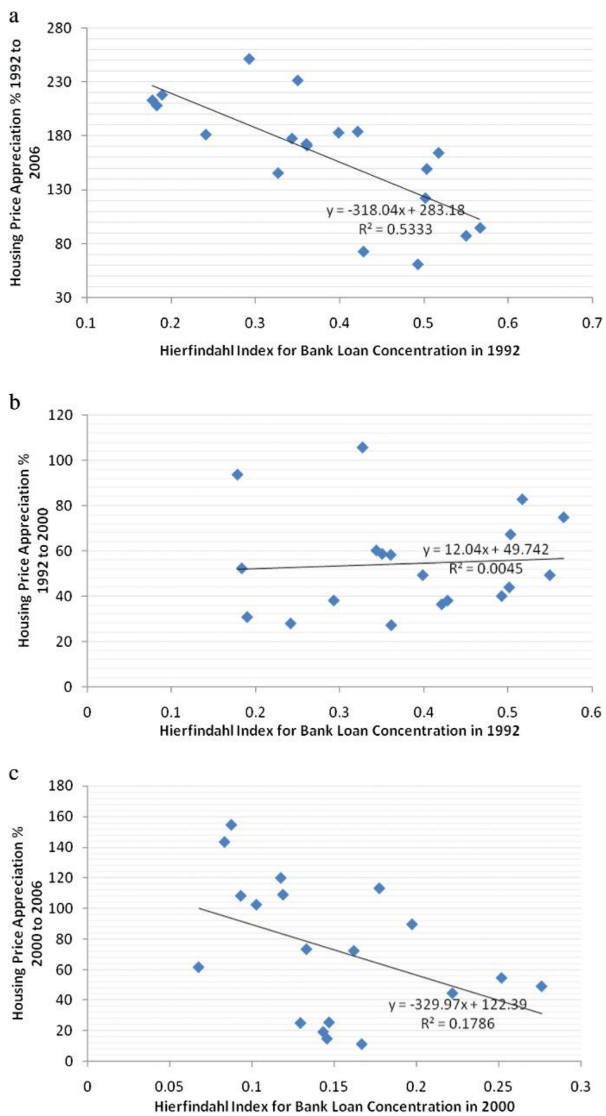

Figure 3. House-price appreciation and lender concentration across Case-Shiller MSAs.